*American Economic Journal: Applied Economics 2021, 13(1): 170–197*
*https://doi.org/10.1257/app.20190165*

# A Dose of Managed Care: Controlling Drug Spending in Medicaid†

*By* David Dranove, Christopher Ody, and Amanda Starc*

*We study the effect of privatizing Medicaid drug benefits on drug prices and utilization. Drug spending would decrease by 21.3 percent if private insurers administered all drug benefits. One-third of the decrease is driven by private insurers' ability to negotiate prices with pharmacies. The remaining two-thirds is driven by the greater use of lower cost drugs, such as generics, and is only realized in states that give private insurers the flexibility to design drug benefits. Privatization does not reduce prescriptions per enrollee and spending cuts are smaller for drugs that lower medical spending.* (*JEL* G22, H51, I11, I13, I18, I38, L65)

Rising healthcare spending—a key concern for both American households and policymakers—is driven by both prices and use. While there is no consensus about how best to control spending, rising drug prices are of particular concern. In this paper, we document the causal effect of drug procurement institutions on outcomes, including prices. We compare public, state-administered programs and private insurers. The comparison is policy-relevant: public insurance programs increasingly finance and regulate competing private insurance plans rather than provide coverage directly.

Advocates of public administration argue that public insurers can set low prices unilaterally. However, public insurers often reimburse providers based on expected costs and cover nearly all services. As a result, they often set prices too high for low value services and do not encourage the use of high-value services. By contrast, innovation in the private insurance market has increased the use of "managed care." Advocates of privatization argue that private insurers can selectively cover services and negotiate lower prices as a result. We look inside the "black box" of managed

* Dranove: Kellogg School of Management, Northwestern University, 2211 Campus Drive, Evanston, IL 60208, and NBER (email: d-dranove@kellogg.northwestern.edu); Ody: Analysis Group, Inc., 180 North Stetson Avenue, Suite 2300, Chicago, IL 60601 (email: christopher.ody@analysisgroup.com); Starc: Kellogg School of Management, Northwestern University, 2211 Campus Drive, Evanston, IL 60208, and NBER (email: amanda.starc@kellogg.northwestern.edu). Ilyana Kuziemko was coeditor for this article. This research was performed while Ody worked at Kellogg and reflects the views of the authors and not necessarily the views of Analysis Group. The authors acknowledge the valuable advice and suggestions provided by Meghan Busse, Craig Garthwaite, Ben Jones, Jordan Keener, Victoria Marone, Tina Marsh Dalton, Brad Shapiro, Mark Shepard, Ashley Swanson, and Pietro Tebaldi and would further like to thank workshop participants at the NBER Health Care meetings, Northwestern University, Haverford College, the Midwest Health Economics Conference, ASHEcon, and the University of Chicago for their comments and feedback.

† Go to https://doi.org/10.1257/app.20190165 to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

care to understand what drives potential savings from privatization. Private insurers may be able to steer demand to higher value care, negotiate lower prices, or both.

Our setting is the Medicaid program, which covers low-income Americans. Many state Medicaid programs contract with managed care organizations (MCOs) to provide health insurance benefits to program enrollees. However, state Medicaid programs historically reimbursed pharmacies for drugs based on expected costs. States managed drug benefits separately from medical benefits because they were entitled to large discounts from drug manufacturers for beneficiaries enrolled in publicly administered plans.

In March 2010, the Affordable Care Act (ACA) required drug manufacturers to provide identical discounts for MCO enrollees, making it weakly better for states to privatize Medicaid drug benefits post-ACA. Many states responded by privatizing the drug benefit: the share of prescription spending that was privately administered increased from 11 percent as of the second quarter of 2010 to 58 percent as of the third quarter of 2016. We argue that privatization was a response to the ACA rule change, and exploit variation in whether and when states privatized to identify the causal effect of privatization on prescription drug spending.

Full privatization of a state's Medicaid drug benefit decreases drug spending by 21.3 percent. MCOs negotiate point-of-sale prices with pharmacies for identical drugs that are 8.8 percent lower than pre-privatization levels. There is no change in prescriptions per enrollee. Instead, the remaining reduction in spending is due to greater use of low cost drugs, such as generics. We provide evidence against a number of alternative explanations of our results. Crucially, we find no evidence to suggest that MCOs reduce spending by skimping on quality; the spending changes are smaller for "high-value" drugs that prevent unnecessary hospitalizations. Furthermore, our results are not driven by enrollee selection: the timing of privatization is not coincident with large changes in aggregate enrollment or Medicaid beneficiary characteristics. The savings are larger in states that allow MCOs to create their own formularies—lists of favored drugs—for Medicaid patients.

While a number of studies contrast public and private provision in Medicare (Town and Liu 2003, Curto et al. 2019) and Medicaid (Duggan 2004; Duggan and Hayford 2013; Aizer, Currie, and Moretti 2007; Currie and Fahr 2005; Vabson 2017), researchers rarely observe both prices and quantities of care under both systems. Yet the seminal studies documenting variation in healthcare utilization (e.g., Finkelstein, Gentzkow, and Williams 2016; Fisher et al. 2003) and prices (e.g., Anderson et al. 2003, Cooper et al. 2019) rarely observe variation in procurement institutions. Our paper contributes to the intersection of these literatures. We illustrate how drug procurement institutions affect prices and quantities. Cutler, McClellan, and Newhouse (2000) (in privately funded healthcare) and Duggan and Scott Morton (2010) (in the Medicare program) document that MCOs obtain lower prices than alternative healthcare purchasers. The magnitudes in our analyses are similar to those in Duggan and Scott Morton (2010), despite differences in the institutional setting.

Both reductions in point-of-sale prices for identical drugs and a shift to generic drugs drive our results. Consistent with Curto et al. (2019), who study privatized Medicare, we document meaningful substitution to lower-cost alternatives. Taken together, our results show that private insurers reduce the cost of providing

prescription drugs without harming enrollees. They are better able to steer demand away from high-cost drugs and can negotiate lower prices with pharmacies. The results illustrate that the private sector provides one type of social insurance at lower cost than the public sector. Privatization in our 13 treatment states, which collectively enrolled roughly a third of Medicaid beneficiaries, reduced annual Medicaid prescription drug spending by more than $1.2 billion.

The paper is organized as follows. In Section I, we provide an overview of the relevant features of the Medicaid program, particularly the role of managed care in Medicaid and the structure of drug benefits. Section II provides evidence on the quasi-experimental variation that underlies our analysis and describes the data. Section III describes our empirical strategy. Sections IV and V detail the paper's central results measuring the effect of privatization on drug spending, quantities, point-of-sale price, and rebates. Section VI explores the robustness of our results and includes extensions. Section VII concludes.

## I. Setting

Medicaid is the main health insurer for low-income families in the United States, and is funded with a combination of state and federal dollars. Some aspects of Medicaid are federally regulated, but states have considerable latitude in designing the program. Historically, eligibility was tied to income and/or assets, but also typically limited to infants, children, pregnant women, and the disabled. As a result of the ACA, many states expanded eligibility to all low-income adults. As of December 2016, Medicaid enrolled more than 68 million individuals—roughly one in five Americans. As of 2010, roughly 50 percent of Medicaid enrollees were covered by a MCO responsible for medical spending.[1] Privatization of Medicaid benefits can vary based on an enrollee's location, reason for eligibility (e.g., pregnancy or disability), and benefit type (e.g., primary care or prescription drugs). Existing research suggests that Medicaid MCOs enroll a healthier subset of the Medicaid population (Duggan and Hayford 2013).

Drugs are a small but growing share of Medicaid spending, increasing from roughly 6 percent of spending to roughly 9 percent from 2010 to 2015. Medicaid enrollees in all states receive drug coverage and pay very little for their drugs regardless of whether their benefit is publicly or privately administered (Medicaid and CHIP Payment and Access Commission 2018). Drug manufacturers are obligated to provide discounts, known as "rebates," to state Medicaid programs according to statutory formulas. Essentially, the formula dictates that the per prescription rebates that states receive must be higher than the rebates received by any nonpublicly funded drug purchaser. Historically, drug manufacturers were not required to provide rebates to states or MCOs for drugs purchased by MCOs for Medicaid patients. Mandatory rebates for state drug purchases, but not for MCO purchases, provided a strong incentive for states to retain control of drug spending. In 2010, less than 15 percent of Medicaid drug spending was administered by MCOs. The ACA mandated that

[1] One-third of Medicare enrollees are in a MCO (Duggan, Starc, and Vabson 2016).

drug manufacturers give states rebates on drugs purchased by MCOs for Medicaid patients.[2] Consequently, privatizing drug benefits became weakly more attractive post-ACA than it was pre-ACA.

The result was a dramatic shift of drug benefits administration away from state governments to MCOs. By 2016, 13 states transitioned from public provision to MCO provision of Medicaid drug benefits, and MCOs managed more than 55 percent of Medicaid drug benefits. The increase in the share of Medicaid drug benefits administered by MCOs is more dramatic in the 13 treatment states. As Figure 1 illustrates, in the second quarter of 2010 (the first quarter in our analyses), effectively no drug spending was administered by MCOs in our treatment states. By the third quarter of 2016 (the final quarter of our analyses), more than 80 percent of drug spending was administered by MCOs.[3] Figure 1 also presents each state's privatization date, and notes the relationship between a state's privatization date and whether and when the state expanded Medicaid. States privatized in a staggered way, with only North Dakota and Indiana privatizing Medicaid drug benefits and expanding Medicaid concurrently.

We next describe how privatization could reduce drug spending. We define drug spending $c_{it}$ for enrollee $i$ in quarter $t$ as

$$c_{it} = \sum_h \sum_d p_{hdt} q_{ihdt}, \tag{1}$$

where $d$ represents an individual drug and $h$ represents an individual pharmacy. MCOs could alter quantities $q_{ihdt}$ and/or prices $p_{hdt}$, which we address in turn.

MCOs could reduce drug prices; online Appendix Figure A.11 is a diagram showing how prices are determined for both public and private Medicaid drug administration. The actual amount that a state or MCO spends on a drug is $p_{hdt}$, and is equal to the price paid to pharmacies ($\varphi_{hdt}$) minus any rebates from drug manufacturers ($\rho_{dt}$): $p_{hdt} = \varphi_{hdt} - \rho_{dt}$. Pharmacy prices are also known as "point-of-sale prices," and are the primary focus of our analysis. Point-of-sale prices and rebates are set separately using different mechanisms. Under public administration, federal regulations mandate that point-of sale prices equal a pharmacy's expected average cost. States reimburse pharmacies the sum of an estimated ingredient cost and a dispensing fee, both of which are administratively determined.

In contrast to the highly regulated government procurement system, MCOs can selectively contract with pharmacies and bargain over point-of-sale prices. Ingredient costs and dispensing fees are fungible from the perspective of both the MCO and the pharmacy, so we consider the overall point-of-sale price. The MCO's bargaining position depends on how hard it is to exclude pharmacies from their network and is largely determined by competition for enrollees (Ho and Lee 2017) and political constraints, including advocacy and legal action by excluded pharmacies. The pharmacy's bargaining position depends on the ability of the MCO to

[2] States and MCOs are free to negotiate supplemental rebates with drug manufacturers both pre- and post-ACA; empirically supplemental rebates are minimal.

[3] Delaware, Utah, and Iowa had very limited experiences with privatized drug benefits prior to the broader adoption decision that we study.

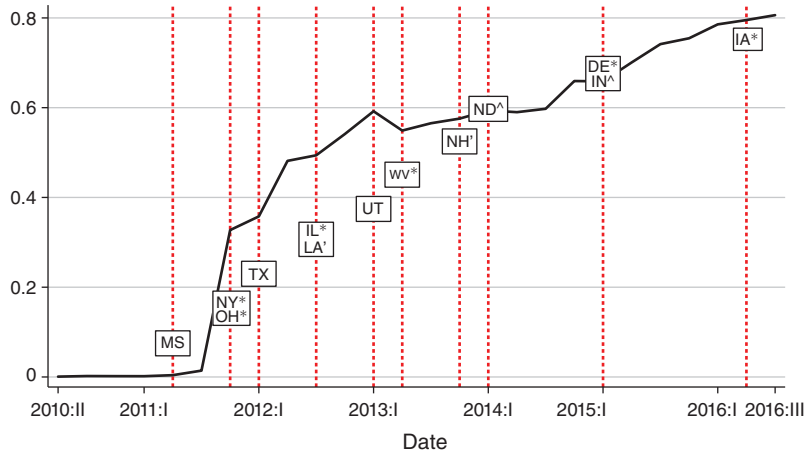


Figure 1. Share of Drug Spending Administered by MCOs in 13 Treatment States

*Notes:* Dashed red lines and state abbreviations give privatization dates. The *y*-axis shows the share of drug spending by MCOs in the 13 treatment states. Medicaid expansion dates are reflected with the following symbols: * (2014:I); ^ (concurrent with privatization); or ' (some other date).

steer consumers away to (potentially lower cost) competitor pharmacies if an agreement is not reached.[4] Anecdotally, exclusion of high-cost independent pharmacies is important for cost control (Ross 2017).[5]

Two institutional factors suggest that MCOs will be able to negotiate lower point-of-sale prices. First, the per prescription dispensing fees are typically set by law to be higher for Medicaid than for private insurers (Cardona 2011). Second, Medicaid's estimated ingredient costs are often inflated well above actual drug acquisition costs (Alpert, Duggan, and Hellerstein 2013).[6] As a result, reimbursed point-of-sale prices may greatly exceed pharmacies' marginal costs, and a MCO may be able to negotiate lower point-of-sale prices than previously administered rates. While these two facts are specific to Medicaid, the incentives that contribute to them are likely of broader relevance to public payers.[7] Finally, total prices also depend on rebates from drug manufacturers ($\rho_{dt}$). Drug manufacturers are required to pay state governments large, administratively determined rebates on all Medicaid drug spending. In addition, both MCOs and states have the option to negotiate with drug manufacturers to obtain additional rebates. Given the change in the underlying

[4] MCOs often set binding point-of-sale price ceilings above which they will not reimburse a pharmacy, leaving the pharmacy to decide between accepting the price ceiling as reimbursement and filling the prescription or not filling the prescription.

[5] This could harm enrollee health if reduced access to pharmacies reduced adherence; we will explore the effect of privatization on quantities in our empirical work.

[6] Within the context of Medicaid drug benefits, they show that cost-based reimbursement has led to "perverse competition," in which manufacturers inflate estimated ingredient costs and pharmacies preference drugs with particularly inflated estimated costs.

[7] When public payers reimburse based on a drug's (or procedure's) expected average costs and do not steer consumers, they eliminate provider incentives to offer high-value drugs. Furthermore, the average cost of drugs is often driven to some extent by reimbursement levels, which can distort average costs away from competitive levels.

price setting mechanism, both rebates and point-of-sale prices could increase or decrease post-privatization.

MCOs could also reduce the quantity of drugs consumed. We consider three factors affecting quantities. First, MCOs could reduce the provision of drugs overall, leading to a lower $q_{ihdt}$ for all $d$. Second, MCOs could alter the mix of drugs consumed, favoring more cost-effective drugs such as generics. Although Medicaid rules limit enrollee cost-sharing and require some degree of access to almost all drugs, MCOs can affect overall drug usage and the mix of drugs consumed through the use of drug formularies.[8] Third, healthy, lower-cost enrollees could select into MCOs.[9] We perform our analyses at the state level: by aggregating over MCO and non-MCO enrollees within a state, we eliminate concerns that the subset of enrollees within a state that are enrolled in managed care may be nonrandom. To alleviate concerns that privatization might lead to changes in the composition of a state's Medicaid population, we show that there are no changes in aggregate enrollment or enrollees' characteristics that are coincident with privatization.[10]

## II. Data

### A. *Drug Benefit Privatization Decisions*

We divide states into three categories at any given point in time: (i) state-administered medical, state-administered drug (public/public); (ii) private MCO-administered medical, state-administered drug (private/public); and (iii) private MCO-administered medical, private MCO-administered drug (private/private). There are no instances of state-administered medical benefits and MCO-administered drug benefits; in private/private states, medical and drug benefits are both administered by the same MCO. After the implementation of the ACA, there are nine potential transitions as described in Table 1.[11]

In our baseline analysis, we use the 16 states that are public/public or private/public in both the pre-period and the post-period as control states, and compare them against the 13 states that transitioned to private/private in the post-period. As of 2010, 16 percent of Medicaid enrollees were in the control states and 33 percent were in the treatment states. We exclude the 22 states that remained private/private.[12] By the end of our sample period, 47 of 51 states are either public/public or private/private. States rarely exclude drugs from MCO contracts in

[8] Formularies enumerate the drugs that Medicaid enrollees can take without restrictions, and the drugs that are only available in limited quantities or with documentation of medical necessity.

[9] Private insurers have incentives to cherry-pick healthy enrollees. Screening incentives affect contract design in Medicare Part D (Lavetti and Simon 2018, Carey 2017), Medicaid (Kuziemko, Meckel, and Rossin-Slater 2018), and the health insurance marketplaces (Geruso, Layton, and Prinz 2019).

[10] We test for, but do not find, heterogeneity in the main effect by risk adjustment regime in online Appendix Table A.5.

[11] Online Appendix Table A.16 identifies which states undergo each of the nine transition types. All data is cited below (Centers for Medicare and Medicaid Services 2002–2015, 2013–2018, 2010–2017, 2010–2014; US Food and Drug Administration 2000–2016a, 2000–2016b; Healthcare Cost and Utilization Project 2006–2015; National Association of Insurance Commissioners 2007–2014; Blewett et al. 2018; Ruggles et al. 2018).

[12] As online Appendix Figure A.12 illustrates, the percent of drug benefits privately administered in these states grows from 26 percent to 64 percent over our sample period.

Table 1—State Transition Types

| | | $t = T$ | | |
|---|---|---|---|---|
| (medical/drug) | | public/public | private/public | private/private |
| $t = 0$ | public/public | *11* | *0* | **5** |
| | private/public | *1* | *4* | **8** |
| | private/private | 0 | 0 | 22 |

*Notes:* States are classified as private if a significant number of Medicaid enrollees are in MCOs that bear financial risk for enrollees' medical spending. The cells for treatment states contain **bold numbers**. The cells for control states contain *italicized numbers*.

the post-period, and therefore a state's decision about whether to privatize medical benefit predicts the state's decision for a drug benefit.

Eight of the 13 states in our sample transitioned from private/public to private/private, likely reflecting a response to the ACA rule change rather than a sudden change in preferences about drug benefit privatization. Only five of the 16 states that began as public/public privatized their drug benefits, and all of those states did so as part of a broader decision to privatize medical benefits. For these states, drug privatization is likely incidental to the medical privatization decision, alleviating concerns that the transition might be a function of unobservable trends in drug costs.

## B. *Data*

For each National Drug Code (NDC) in each state and in each quarter, the Medicaid State Drug Utilization Data contain the number of prescriptions filled as well as point-of-sale (i.e., prerebate) prices.[13] Since the passage of the ACA, MCO drug purchases have been collected and reported separately within the same dataset. Therefore, our analysis covers the second quarter of 2010 (the first quarter for which MCO drug utilization was reported), through the third quarter of 2016. Given that reimbursements and rebates rely on these data, state governments, the federal government, and drug manufacturers have strong incentives to ensure that they are accurate.

We obtain additional information from a number of sources. We collect data on the timing of Medicaid expansions and obtain enrollment data from the Medicaid Enrollment Reports. We merge the Medicaid State Drug Utilization data with current and historical versions of the Drugs at the FDA database and the FDA's National Drug Code (NDC) Directory, which enables us to determine each NDC's active ingredient and whether it is branded or generic.[14]

[13] An NDC is a unique identifier for human drugs in the United States. It uniquely identifies the labeler (roughly, the pharmaceutical manufacturer); the specific strength, dosage form (i.e, capsule, tablet, liquid) and formulation; and the package size and type. A given drug as defined by its brand name or generic name (e.g., atorvastatin calcium for treatment of cholesterolemia, or the original atorvastatin brand Lipitor) may have many associated NDCs.

[14] In analyses that rely upon FDA data, we exclude nonmatching NDCs, which account for roughly 3 percent of revenues and often are nonprescription drugs.

We construct several dependent variables. *Share MCO* is the share of Medicaid drug spending in a state-quarter that is administered by MCOs. *Drug spending per enrollee* is drug spending per quarter divided by Medicaid enrollment, and is equal to *prescriptions per enrollee* times (point-of-sale) *price per prescription*.[15] *Drug utilization per enrollee* holds point-of-sale prices fixed: specifically, we replace the NDC-state-quarter point-of-sale price per prescription with the average point-of-sale price per prescription for that NDC, and then recalculate spending at the new (average) prices. To measure the use of inexpensive generics, we create three variables: *generic penetration* is the share of prescriptions that are filled with a generic; *generic efficiency* is the share of prescriptions filled with a generic when a generic is available; and *generic accessibility* is the share of prescriptions that are for drugs for which a generic is available. *Generic penetration* is equal to *generic efficiency* times *generic accessibility*. Finally, we create *share high offset*, which is the share of drug spending on "Category 1" drugs, as classified by Chandra, Gruber, and McKnight (2010).[16] If these drugs are not taken, a serious event, such as hospitalization, is likely to occur within the next six months. For example, drugs that treat asthma, diabetes, high cholesterol, and high blood pressure are high offset drugs.

Table 2 compares summary statistics for the second quarter of 2010 in treatment and control states.[17] *Drug spending per enrollee* and *drug utilization per enrollee* are economically similar and statistically indistinguishable for the treatment and control states. However, there is evidence that the treatment and control states differ along several dimensions. For example, utilization of high offset drugs is 3.6 percentage points lower in treatment states than in control states. This difference suggests that there is more room to increase this measure of efficiency in the states that privatized drug benefits. Finally, treatment states were more likely than control states to expand Medicaid under the ACA. Section IV describes additional robustness checks showing that the results are not driven by the Medicaid expansions; we also present analyses both excluding and including a control for the Medicaid expansion, which does not affect any of the results.

## III. Empirical Strategy

We conduct most of the analyses at the state-quarter level. For state $s$ in quarter $t$ (e.g., quarter 2 of 2010), let $Y_{st}$ be one of the dependent variables of interest, $\alpha_s$ be a state-specific fixed effect, $\tau_t$ be a quarter fixed effect, $priv_{st}$ be an indicator for whether a state has begun to privatize its Medicaid drug

[15] Medicare-Medicaid dual eligibles receive drug coverage from Medicare Part D, and therefore their prescription drug use is not included in the Medicaid State Utilization Data. When constructing Medicaid drug spending per enrollee, we exclude Medicare-Medicaid dual eligibles from the denominator. The exact timing of the historical reports that we were able to locate varies across years. We interpolate the enrollment data to make it available for each quarter.

[16] To ensure this measure captures differences in utilization rather than differences in pricing, we replace the NDC-state-quarter point-of-sale price per prescription with the average point-of-sale price per prescription for that NDC, and then recalculate spending at the new (average) prices.

[17] Online Appendix Figure A.13 presents histograms of our key dependent variables for treatment and control states. Online Appendix Figure A.14 illustrates there is no geographic clustering of our treatment and control states.

Table 2—Summary Statistics

| | Treatment | | Control | | Difference | |
|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD |
| *Medicaid drug characteristics* | | | | | | |
| Share MCO | 0.0016 | 0.0051 | 0.0000 | 0.0001 | 0.0016 | 0.0013 |
| Drug spending per enrollee | 209 | 57 | 212 | 64 | −3 | 23 |
| Prescriptions per enrollee | 3.162 | 0.892 | 2.947 | 0.800 | 0.2156 | 0.314 |
| Price per prescription | 66.44 | 8.22 | 72.53 | 15.42 | −6.08 | 4.75 |
| Price per generic prescription | 19.59 | 4.86 | 22.26 | 11.55 | −2.66 | 3.43 |
| Price per branded (no gen comp) prescription | 248.48 | 25.62 | 265.18 | 66.49 | −16.69 | 19.57 |
| Price per branded (w/gen comp) prescription | 102.45 | 14.15 | 114.37 | 21.29 | −11.93 | 6.89 |
| Drug utilization per enrollee | 251.84 | 74.21 | 239.00 | 69.70 | 13 | 26.79 |
| Generic penetration | 0.6611 | 0.0375 | 0.6679 | 0.0315 | −0.0069 | 0.0128 |
| Generic efficiency | 0.7883 | 0.0249 | 0.7930 | 0.0192 | −0.0047 | 0.0082 |
| Generic accessibility | 0.8381 | 0.0284 | 0.8419 | 0.0229 | −0.0038 | 0.0095 |
| Share high offset | 0.4190 | 0.0383 | 0.4546 | 0.0400 | −0.0356 | 0.0146 |
| *Overall Medicaid characteristics* | | | | | | |
| Share Black | 0.2110 | 0.1735 | 0.1422 | 0.1348 | 0.0688 | 0.0572 |
| Share Hispanic | 0.1264 | 0.1435 | 0.0752 | 0.0766 | 0.0512 | 0.0416 |
| Share disabled | 0.1496 | 0.0422 | 0.1526 | 0.0319 | −0.003 | 0.0138 |
| Share dual | 0.1589 | 0.0378 | 0.1748 | 0.0478 | −0.0159 | 0.0163 |
| Share in medical MCO | 0.3069 | 0.3194 | 0.1785 | 0.3132 | 0.1283 | 0.1180 |
| *State characteristics* | | | | | | |
| Share with Medicaid | 0.1641 | 0.0544 | 0.1562 | 0.0429 | 0.0078 | 0.0181 |
| Expands Medicaid? | 0.7692 | 0.4385 | 0.3125 | 0.4787 | 0.4567 | 0.1722 |
| Has Republican governor | 0.4615 | 0.5189 | 0.4375 | 0.5123 | 0.024 | 0.1924 |
| Republican share of legislature | 0.4682 | 0.1220 | 0.5065 | 0.1571 | −0.0383 | 0.0532 |
| Observations | 13 | | 16 | | | |

*Note:* Unit of observation is a state in the second quarter of 2010.

benefit, and $Medicaid\ expansion_{st}$ be an indicator for whether the state has expanded Medicaid under the ACA. Our estimating equation is

$$Y_{st} = \alpha_s + \tau_t + \beta priv_{st} + \lambda Medicaid\ expansion_{st} + \varepsilon_{st}. \tag{2}$$

To study the timing of the effect, we replace *priv* in equation (2) with indicators for the number of quarters since a state began privatizing drug benefits. More precisely, let $E_s$ be the quarter in which state $s$ privatizes. Let $D_{st}^l = \mathbf{1}(t - E_s = l)$ be an indicator that is equal to one if and only if state $s$ is $l$ quarters from privatization at time $t$.[18] Then, our event study specification is

$$Y_{st} = \alpha_s + \tau_t + \sum_{l=-9}^{9} \beta^l D_{st}^l + \lambda Medicaid\ expansion_{st} + \varepsilon_{st}. \tag{3}$$

We plot these lag/lead coefficients to illustrate the extent of privatization over time and the resulting effect on outcomes of interest. We weigh observations by each

[18] We allow for nine lags and leads. We therefore set $D_{st}^{-9} = \mathbf{1}(t - E_s \leq -9)$ and $D_{st}^{9} = \mathbf{1}(t - E_s \geq 9)$. Furthermore, we set $\beta^{-1} = 0$.

state's Medicaid drug spending in the second quarter of 2010, the first quarter of our analysis, and cluster standard errors by state.

The coefficient on *priv* is the difference-in-differences (DID) effect of privatizing some drug benefits. We also refer to this as the "reduced-form" analysis. When states privatize some drug benefits, they do not switch all drug spending into MCOs.[19] We also create instrumental variable (IV) estimates of the effect of full privatization of drug benefits. To do this, we estimate the "first-stage" relationship between privatizing some drug benefits and *share MCO*. The IV estimates of the effect of *share MCO* on our outcomes of interest rescales the reduced-form estimates by dividing them by the first-stage estimates.

### A. *First Stage*

Figure 2 plots the coefficients on lags and leads of privatization using *share MCO* as the dependent variable. The share of drug spending administered by MCOs is flat prior to privatization, jumps by nearly 60 percent in the quarter after privatization begins, and reaches nearly 80 percent within two years.[20] When states begin privatizing, not all drug spending shifts to MCOs: MCO enrollment is sometimes voluntary and not every eligible or geographic group is shifted into managed care.[21] However, the large point estimate in the first-stage regression suggests that the local average treatment effect that we estimate will be identified off of privatization-induced spending changes from a large share of Medicaid enrollees. Panel A of online Appendix Table A.17 presents estimates of a regression in which we collapse the lags and leads into a pre-post indicator variable. The pooled post-period estimate of the effect of privatization on the share of drug spending administered by MCOs is 60.8 percentage points.[22] The IV estimates will effectively rescale the reduced-form DID estimates by these first-stage estimates (by approximately $1/0.608 = 1.64$).

[19] The enrollees that switch into MCOs are likely a nonrandom subset of enrollees. Specifically, many states move healthier enrollees into MCOs. For this reason, we do not compare spending within a state of those with MCO drug benefits relative to those with state administered drug benefits. Our estimates will represent a local average treatment effect for those switching into MCOs and therefore our results might not generalize to other groups of enrollees.

[20] The estimated coefficients are all near zero prior to privatization, but are not exactly zero for two reasons. First, even if *share MCO* was exactly zero in the pre-period, the coefficient would not be zero because of a general time trend toward greater privatization. Second, as noted above, states such as Delaware, Utah, and Iowa had very limited experiences with privatized drug benefits prior to the broader adoption decision that we study.

[21] Online Appendix Figure A.15 plots the share of drug spending by MCOs in each of our 13 treatment states over time; the vertical dashed line denotes the privatization date. For most states, there is a sharp break in the share of drug spending administered by MCOs at the time of privatization.

[22] In column 2, we include pre-post indicators for states that expanded Medicaid, using the appropriate timing for each state, to control for any changes in the composition of enrollees. The coefficient on *Medicaid expansion* is close to zero, and the coefficient on *priv* is unchanged and still highly significant.

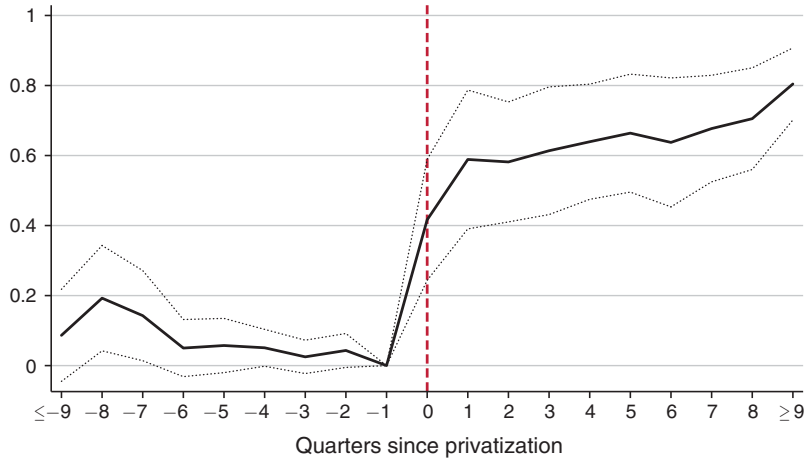


Figure 2. Privatization and *share MCO*

*Notes:* Figure includes state fixed effects, quarter fixed effects, and a post-Medicaid expansion indicator. Observations are weighted by state Medicaid drug spending in the second quarter of 2010. The solid line is point estimates for the effect of quarter pre-post privatization and the dotted lines are the 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state.

## IV. Impact on Drug Use

### A. *Impact on Prescriptions Per Enrollee*, *Price Per Prescription*, *and Spending Per Enrollee*

In this section, we estimate the effect of privatization on prescriptions per enrollee, prices per prescription, and drug spending per enrollee. Panel A of Figure 3 shows that drug consumption, as measured by prescriptions per enrollee, is unchanged in the post-period. Panels B through E of online Appendix Table A.17 present estimates of a regression in which we collapse the lags and leads into a pre-post indicator variable. Column 1 shows the reduced-form estimate of the effect of a post-privatization indicator variable, *priv*. Column 2 presents the IV estimate, while columns 3 and 4 include a post-Medicaid expansion indicator.[23] The pooled specification implies that full privatization would increase *prescriptions per enrollee* by 9 percent, which is not statistically significant.

By contrast, panel B of Figure 3 shows that the average point-of-sale price per prescription is flat before privatization and falls immediately following privatization; full privatization of drug benefits would reduce *price per prescription* by 27.8 percent.[24] This reduction in average price could be due to lower prices for the

[23] Throughout the text, we discuss column 4, which is our preferred estimate in both this table and similar tables in other sections. In all instances, the results are unaffected by the inclusion of an indicator for the Medicaid expansion.

[24] If the effect of privatization on the logged dependent variable is $\beta$, then privatization causes a $100 \times \left(e^{\beta} - 1\right)$ percent change in the dependent variable.

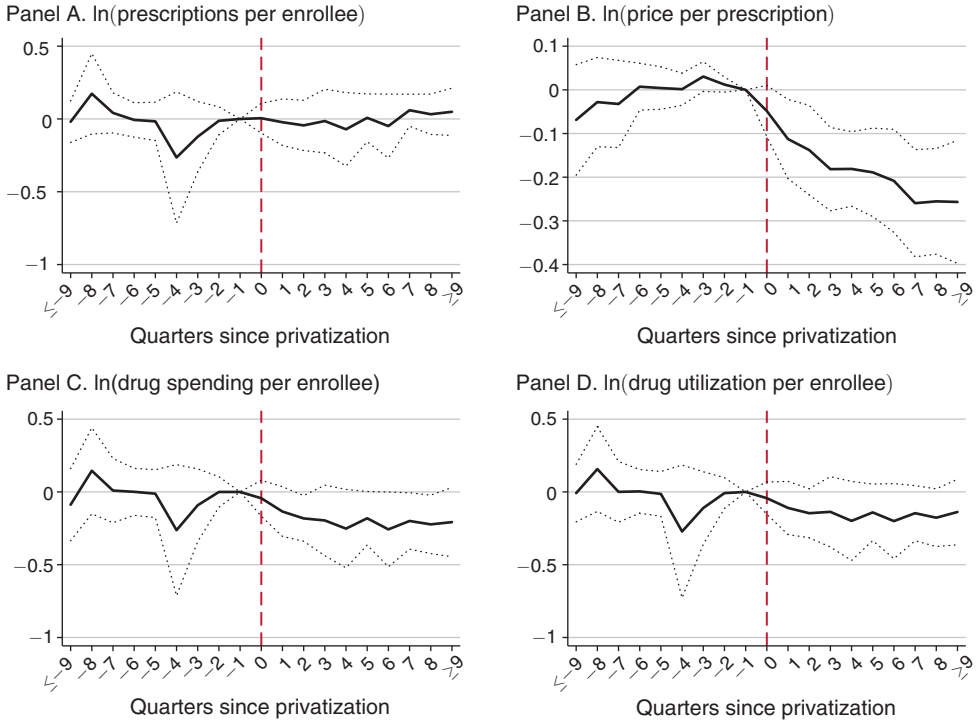


Figure 3. Privatization and Drug Spending

*Notes:* Figure includes state fixed effects, quarter fixed effects, and a post-Medicaid expansion indicator. Observations are weighted by state Medicaid drug spending in the second quarter of 2010. The solid line is point estimates for the effect of quarter pre-post privatization and the dotted lines are the 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state.

same drugs or a shift toward less expensive drugs, or both. We will decompose the point-of-sale price decrease into these two mechanisms.

Our figures show that the effect of privatization grows over time, which is sensible given market institutions. Patients must visit physicians and fill prescriptions; inertia in drug choice among consumers may also delay the effect of the policy change. Furthermore, prices are typically renegotiated on an annual or biannual basis, while we use the quarter as unit of analysis in our study.

We next present the net effect of flat drug consumption and lower point-of-sale drug prices on drug spending per enrollee. Panel C of Figure 3 plots the coefficients of lags and leads of privatization on the log of *drug spending per enrollee*. While *drug spending per enrollee* fluctuates prior to privatization, it drops following privatization: full privatization of drug benefits would reduce *drug spending per enrollee* by 21.4 percent. In our 13 treated states, which account for roughly one-third of Medicaid enrollees, privatization of roughly 80 percent of Medicaid drug spending reduces annual drug spending by about 1.2 billion dollars.

Panel D of Figure 3 illustrates that privatization reduces *drug utilization per enrollee* (spending holding prices fixed at their national averages). The point estimate implies that full privatization reduces *drug utilization per enrollee* by 14.1 percent

but this is not statistically distinguishable from zero. The effect of full privatization on *drug utilization per enrollee* is about two-thirds of the effect on *drug spending per enrollee*, suggesting that roughly one-third of the decrease in spending is due to price changes. We explore the effect of privatization on within-drug point-of-sale prices in Section V.

### B. *Impact on Generic Penetration*

Here, we estimate the effect of privatization on the composition of consumed drugs. Panel A of Figure 4 highlights one potential driver of the decline in point-of-sale prices: substitution from branded to generic drugs. We plot the coefficients on lags and leads of the privatization decision on *generic penetration* (share of prescriptions that are filled with a generic). After a decrease six quarters prior to privatization, generic penetration remains flat for the next six quarters. Generic penetration increases steadily beginning at the time of privatization; two years after privatization, it has risen by roughly 5 percentage points. Full privatization causes generic penetration to increase by 7.73 percentage points: an 11.7 percent increase on a base of 66 percent.[25]

Panel B of Figure 4 presents the effect of privatization on *generic efficiency* (share of prescriptions that are filled with a generic when a generic is available). Panel C of Figure 4 presents the effect of privatization on *generic accessibility* (share of prescriptions for drugs for which a generic is available). Full privatization would decrease the share of prescriptions filled for molecules for which no generic is available by 5.26 percentage points, or 29 percent (on a pre-privatization mean of 18.6 percent). These results imply that roughly two-thirds of the increase in generic penetration results from across-molecule shifts and roughly one-third is a result of within-molecule shifts from branded to generic.[26]

A back-of-the-envelope calculation, described in online Appendix Table A.19, suggests that increased use of generic and generic-accessible drugs decreases drug spending by 8.9 percent. Price decreases for identical 9-digit NDCs and across-molecule but within drug class substitution from branded drugs to generics are the two most important mechanisms driving our results.[27]

One concern is that increases in generic accessibility could be driven by patent expiration, rather than molecular substitution: differences across states in the types of drugs used by Medicaid patients combined with patent expiration could lead to changes in generic accessibility. To rule out this concern, we create *simulated generic accessibility*, which is how generic accessibility would have changed over time (as a result of patent expiration) if molecule-level drug shares remained fixed at the second quarter of 2010 levels. Panel D of Figure 4 illustrates that privatization does not affect this variable.

[25] Pooled results for this section are presented in online Appendix Table A.18.

[26] The share of prescriptions for branded drugs with no generic equivalent available decreased by 5.26 percentage points. The share of prescriptions for generics increased by 7.73 percentage points. It follows that the share of prescriptions for branded drugs for which a generic is available decreased by 2.47 percentage points.

[27] Online Appendix Table A.20 presents this decomposition.

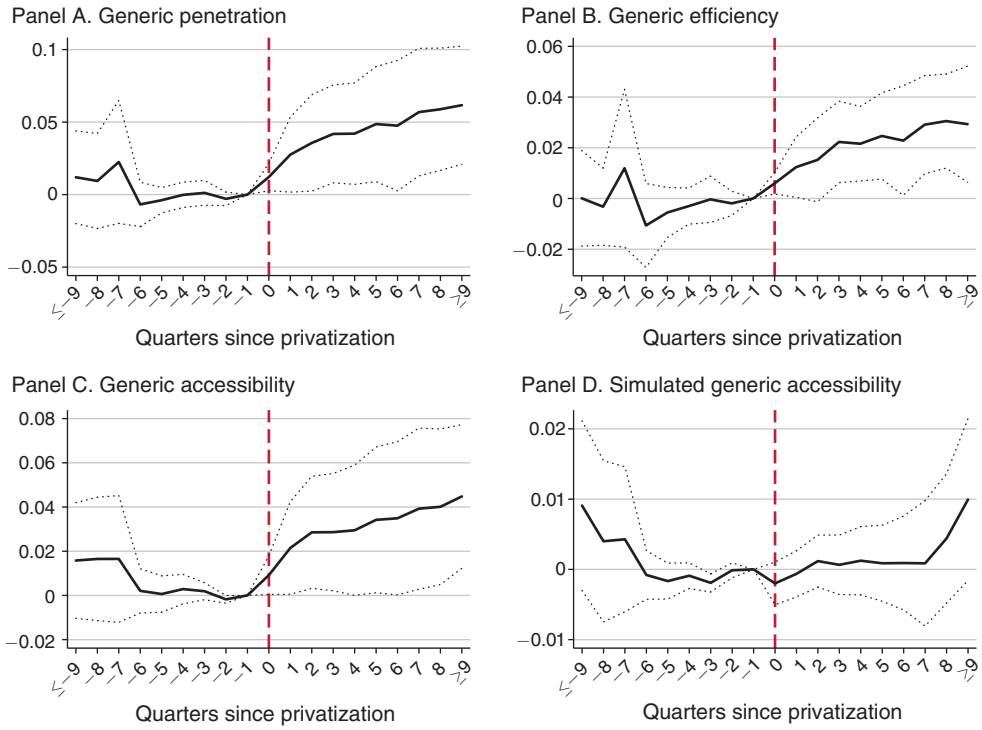


FIGURE 4. PRIVATIZATION AND GENERICS

*Notes:* Figure includes state fixed effects, quarter fixed effects, and a post-Medicaid expansion indicator. Observations are weighted by state Medicaid drug spending in the second quarter of 2010. The solid line is point estimates for the effect of quarter pre-post privatization and the dotted lines are the 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state.

## C. *Impact on Use of High Offset Drugs*

MCOs reduce drug spending; however, if this reduction is achieved by spending less on high-value drugs that offset other spending, overall costs could increase. Figure 5 plots the effect of lags and leads of privatization on *share high offset* (share of drug utilization on drugs likely to prevent immediate adverse health consequences). There is an immediate jump of roughly 3 percent in the share of spending for high offset drugs post-privatization. In panel C of online Appendix Table A.21, we present pooled versions of this analysis; we find that full privatization causes the share of utilization for high offset drugs to increase by 5.44 percentage points. Prior to privatization of drug benefits, MCOs had incentives to promote the use of high offset drugs and would have been relatively indifferent about the use of low offset drugs. Privatization might enhance the MCO's ability to promote high offset drugs while discouraging use of branded drugs when possible. Panels A and B of online Appendix Table A.21 illustrate that there is no evidence of a change in the per enrollee utilization of high offset drugs after privatization. Instead, utilization of all other drugs decreases. Lower utilization—which holds

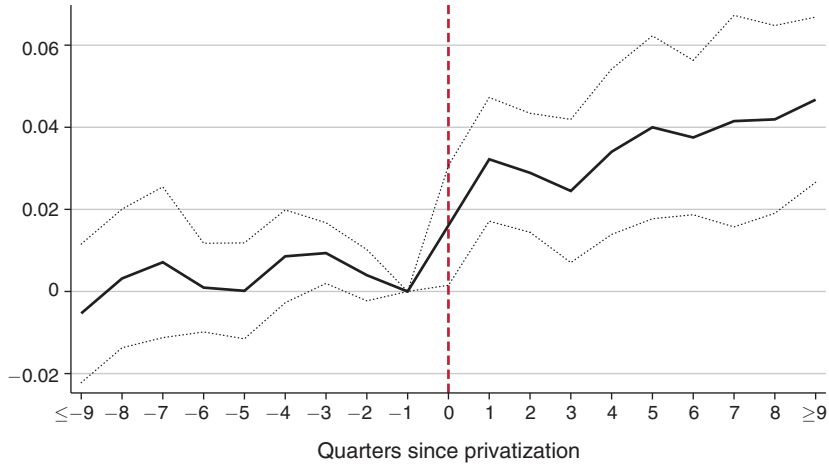


FIGURE 5. PRIVATIZATION AND *share high offset*

*Notes:* Figure includes state fixed effects, quarter fixed effects, and a post-Medicaid expansion indicator. Observations are weighted by state Medicaid drug spending in the second quarter of 2010. The solid line is point estimates for the effect of quarter pre-post privatization and the dotted lines are the 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state.

prices constant—of nonhigh offset drugs reduced spending by roughly 12.9 percent.[28] Table A.22, Figure A.16, and Figure A.17 in the online Appendix reestimate many of the paper's central analyses separately for high offset and nonhigh offset drugs.

MCOs have strong incentives to keep enrollees using high offset drugs. In online Appendix Table A.23, we find more muted changes for drugs used by vulnerable populations.[29] In Table A.23, we find more muted changes for drugs in Medicare's six protected drug classes and among drugs with orphan drug status. Finally, a related concern is that MCOs my be exposing enrollees to substantially more financial risk through higher cost sharing; Appendices A and G do not find any evidence of increased cost sharing, either in general or for specific types of drugs that are clinically beneficial or for which cost sharing might be particularly likely to deter use.

[28] Privatization decreases utilization for nonhigh offset drugs, which account for roughly 60 percent of spending, by 21.5 percent.

[29] We adopt two definitions for drugs that are used by vulnerable populations. The first definition is Medicare's six protected drug classes; Medicare Part D plans are required to provide access to substantially all drugs within these classes. These classes were chosen because of concerns about imperfect substitutability of drugs within these classes and because of concerns that insurers would otherwise avoid covering these drugs to screen out high cost populations. These six protected classes are anticonvulsants, antidepressants, antineoplastics, antipsychotics, antiretrovirals, and immunosuppressants. The second definition is the Food and Drug Administration's orphan drug status. Orphan drugs treat small, often high-cost consumer segments, and therefore insurers may be able to screen out high-cost populations by not covering these drugs.

## V. Impact on Point-of-Sale Prices

In this section, we estimate the effect of privatization on point-of-sale prices for identical drugs, as measured by a 9-digit NDC code.

For NDC $d$ in state $s$ in quarter $t$, let $\alpha_{ds}$ be an NDC-state-specific fixed effect, $\tau_{dt}$ be an NDC-quarter fixed effect, $priv_{st}$ be an indicator for whether a state has begun to privatize its Medicaid drug benefit, and $Medicaid\ expansion_{st}$ be a post-Medicaid expansion indicator. Our estimating equation is given by

$$(4) \qquad \ln(price\ per\ prescription_{dst}) = \alpha_{ds} + \tau_{dt} + \beta priv_{st} + \lambda Medicaid\ expansion_{st} + \varepsilon_{dst}.$$

Drug and time fixed effects control for pricing differentials at a very granular level.

Figure 6 presents leads and lags showing the movement of prices before and after privatization: prices are relatively flat until privatization and then drop by 5.3 percent after privatization, consistent with our previous results.[30] Full privatization reduces prices by 8.4 percent.[31] Because the states' administratively set prices do not vary across pharmacies within a state, the results imply that bargaining lowers prices for at least some pharmacies.

Our data does not allow us to track utilization at the individual pharmacy location level. As a result, one concern could be that lower point-of-sale reimbursements are associated with reduced access to pharmacies or pharmacy exit (Starc and Swanson 2018). We note that Medicaid network adequacy rules require that MCOs maintain "sufficient" networks with respect to both distance and driving time for beneficiaries. Furthermore, in unreported specifications we find no evidence of systematic pharmacy exit in response to the policy change.

*Rebates*.—Total Medicaid spending on drugs depends on both point-of-sale prices and rebates from drug manufacturers. To examine whether privatization affects rebates, we collected supplemental data from the Medicaid Financial Management Report, which provide state-year level summary information on all Medicaid expenditures. On average, rebates reduce Medicaid drug spending by 50 percent. Given that the statutory rebates in the Medicaid program exceed those negotiated by private payers in Medicare Part D (Levinson 2015), we expect the magnitude of any additional rebates to be small. In fact, only

[30] Appendix A presents additional NDC-level analyses. Pooled regressions are presented in online Appendix Table A.1. Columns 1 and 2 exclude all NDC-specific fixed effects. Consistent with previous results, the estimates are meaningfully different once these fixed effects are included. The preferred IV regression with all controls in is presented in column 6. In online Appendix Table A.2, we find that pharmacy reimbursements from Medicaid decrease substantially for drugs that are neither generics nor high offset, while reimbursements for generics remain similar. The effect of privatization on the prices of nongeneric drugs that are high offset is more mixed.

[31] This is similar but not identical to the decrease in spending from price changes in Section A. Factors such as differences in weighting, the level of aggregation, and the controls included probably contribute to the (small) difference.

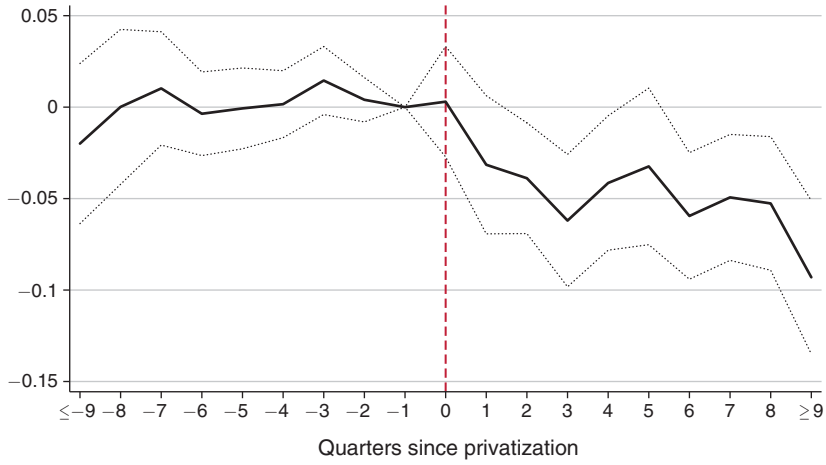


Figure 6. Privatization and ln(*price per prescription*) (within Drug Analysis)

*Notes:* Unit of observation is the NDC-state-quarter. Includes NDC-state fixed effects, NDC-quarter fixed effects, and a post-Medicaid expansion indicator. Observations are weighted by drug spending for each NDC-state combination in the first quarter in which it appears in the data. The solid line is point estimates for the effect of quarter pre-post privatization and the dotted lines are the 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state.

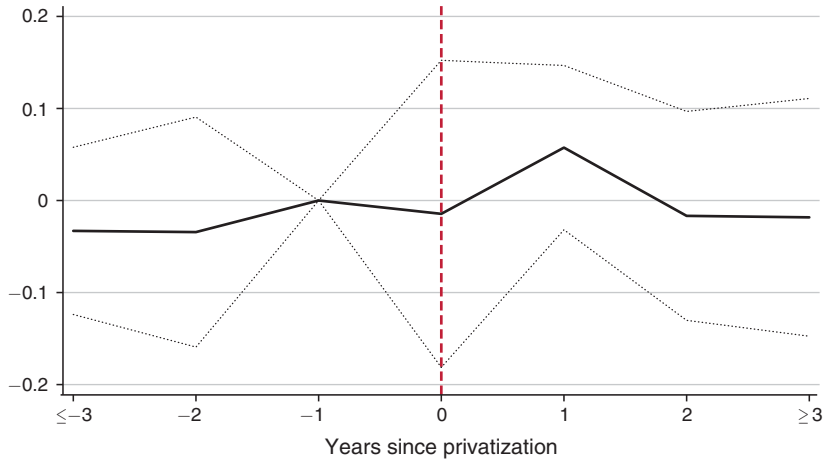


Figure 7. Privatization and *rebate share*

*Notes:* Figure includes state fixed effects, year fixed effects, and a post-Medicaid expansion indicator. Observations are weighted by state Medicaid drug spending in quarters 2 through 4 of 2010. Point estimates for the effect of year pre-post privatization are presented in the dark line, and a 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state, is presented with the dotted lines.

8 percent of rebates received by our sample states in 2010 resulted from negotiations with drug manufacturers; the other 92 percent were classified as federally mandated.

We define *rebate share* as the share of point-of-sale drug spending that is returned in rebates. The numerator is constructed from the Medicaid Financial Management Reports, while the denominator is constructed from the Medicaid State Drug Utilization Data. Figure 7 and online Appendix Table A.24 illustrate that privatization does not affect rebates. Because *rebate share* is constant over time and prices and spending are measured in logs rather than levels, proportional adjustments do not affect those analyses. While we cannot rule out the possibility that privatization affects rebates for some specific drugs, these results suggest that lower point-of-sale prices are not, on average, being offset by lower rebates.

## VI. Robustness and Extensions

### A. *Heterogeneity*

This section summarizes a number of analyses that examine whether there is heterogeneity in the effect of privatization based on state characteristics. Most importantly, we document that the effect of privatization is stronger in states in which MCOs are allowed to design their own formularies. Formularies are an important tool for steering consumers, so we anticipate larger effects in the states in which MCOs have some control over formularies.

Figure 8 presents the effect of privatization on a number of separate dependent variables based upon whether states allow MCOs to design their own formularies.[32] Panel A illustrates that the effect of *drug spending per enrollee* is effectively zero in states in which MCOs cannot design their own formulary. Panel B illustrates that the effect of privatization on *price per prescription* is stronger in states in which MCOs can design their own formularies, but that prices decrease in all states. Additional results, presented in online Appendix Table A.6, show that differences in point-of-sale prices for identical drugs do not vary according to whether states allow MCOs to design formularies. Therefore, we conclude that bargaining with pharmacies—something MCOs can do regardless of whether they control formulary design—is critical for achieving lower point-of-sale prices. Panel C illustrates that *prescriptions per enrollee* are relatively flat in both samples in the post period; there is some increase in states in which MCOs cannot design formularies. Consistent with our hypotheses in Section I, panel D illustrates that the shift toward generics is stronger in states where the MCO designs the formulary.[33] Therefore, restricting formularies is critical for shifting prescriptions to generics. Online Appendix Table A.5 presents IV analogs to the results in Figure 8 for additional independent variables.

In Appendix B, we study additional sources of heterogeneity in the effect of privatization based on state characteristics.

[32] We collect data from the Medicaid Drug Utilization Review (MDUR) to determine whether MCOs are required to use the state's formulary or whether MCOs design their own formularies.

[33] Even in the states in which the MCOs must use the state's formulary, MCOs shifted utilization to high offset drugs (column 9) and substituted non-bio-equivalent generics for branded drugs; such shifts could not be achieved through formulary design alone.

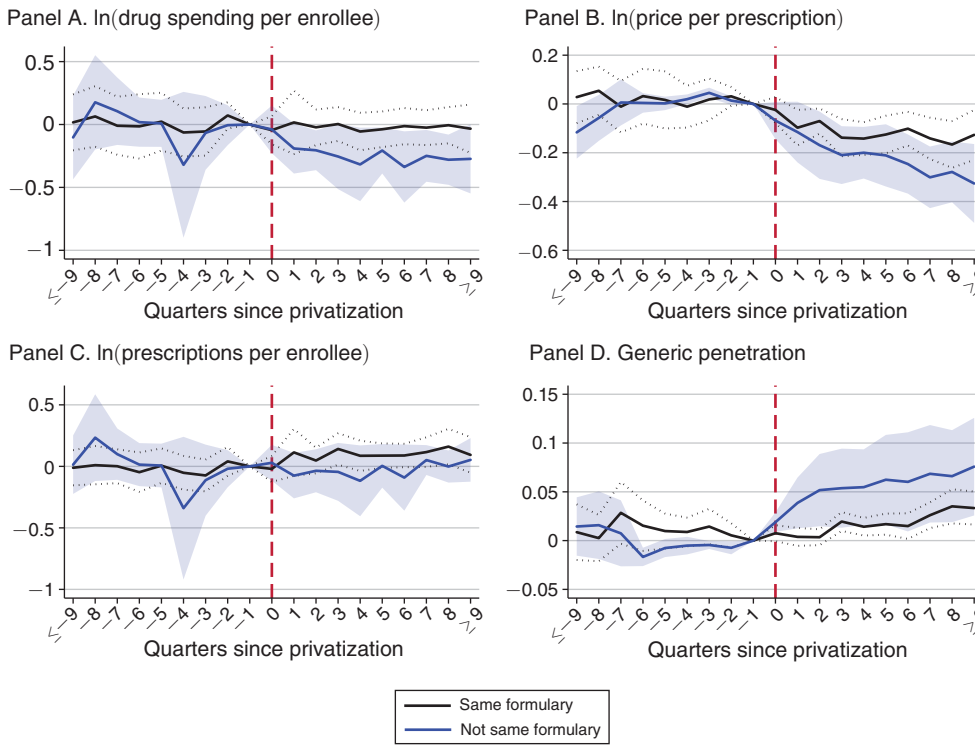


Figure 8. Heterogeneity in Effect of Privatization by whether MCOs Can Design Formularies

*Notes:* Figure includes state fixed effects, quarter fixed effects, and a post-Medicaid expansion indicator. The lags and leads for the two groups of states are from separate regressions; each regression controls for the privatization in the other group of states with a post-privatization indicator. Point estimates for the effect of quarter pre-post privatization by drug type are presented as solid lines and 95 percent confidence intervals for those coefficients, based upon standard errors that are clustered by state, are presented with dotted lines/shading. Formularies are lists of preferred drugs.

## B. *Robustness*

The most serious concern with our analyses is that they may not control adequately for the (potentially heterogeneous) effects of Medicaid expansions and differential selection into Medicaid across states and over time. Online Appendix Table A.8 shows that the results in Sections A, B, and C are unchanged when we include a variety of alternative controls for the Medicaid expansion. Online Appendix Table A.25 shows that the results are robust to excluding states based upon whether they expanded Medicaid. The robustness of these results is unsurprising; privatization is not concurrent with changes in Medicaid enrollment (see online Appendix Figure A.2). Figure 9 and online Appendix Figure A.4, which are discussed in detail in Section C, show that Medicaid enrollee demographics do not change concurrently with privatization.

Online Appendix Table A.26 illustrates that the results are robust to a number of alternative methods of restricting the sample of analysis states. Online Appendix

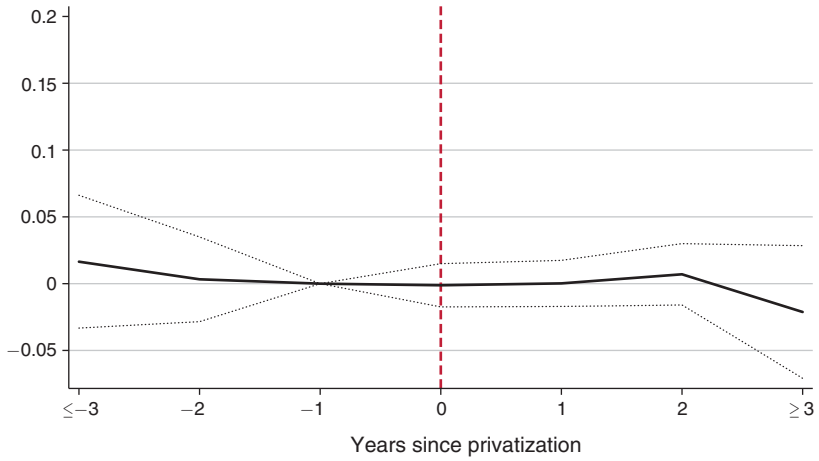


Figure 9. Privatization and Logged

*Notes:* Figure includes state fixed effects, year fixed effects, and a post-Medicaid expansion indicator. The dependent variable is the log of expected Medicaid enrollee drug spending based on age-sex-race/Hispanic status. The average is calculated for individuals in the American Community Survey (ACS) with Medicaid and without Medicare for each state-year. Observations are weighted by the ACS respondents underlying these calculations in 2010 for each state. Further detail on this analysis is in Section C. Point estimates for the effect of year pre-post privatization is presented in the dark line and a 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state, is presented with the dotted lines.

Table A.27 confirms that the results are unchanged by winsorization, controlling for linear time trends and equalizing the number of pre- and post-period quarters for each treatment state. Online Appendix Figure A.18 illustrates that the lags and leads specifications remain unchanged when we restrict the treatment states' data so that the same sample of states identifies all lag and lead parameters. Online Appendix Table A.28 demonstrates that the results are robust to excluding individual treatment states. Online Appendix Table A.29 illustrates that the results are robust to utilizing variation across all states and replacing the privatization variable with *share MCO* (lagged by one quarter).

Table A.32, Figure A.19, and Figure A.20 in the online Appendix illustrate which comparisons between potential treatment and control states drive our results. These are based on suggestions in Goodman-Bacon (2018). Most identifying variation is from comparing outcomes in treatment states after privatization to outcomes in control states. Very little identifying variation is from comparisons within treatment states over time. Our results remain similar under various sample restrictions. We can restrict attention to: (i) privatizing states versus nonprivatizing states, (ii) privatizing states versus other states that will privatize in the future, or (iii) privatizing states versus other states that have privatized in the past. As the event study figures show, the effects of privatization grow over time. Therefore, the point estimates that restrict to (iii) are typically substantially smaller. They do not, however, receive much weight in the two-way fixed effects estimate.

Finally, one concern with privatization is that private firms may profit because of selection. Private firms may skimp on product quality to cherry-pick profitable

patients; no similar incentive exists when a public insurer serves the whole market. The consequences of privatization—lower point-of-sale prices, a shift in spending to high offset drugs, and increased use of generics—are not naturally explained by insurer attempts to offer less generous coverage to cherry pick. We circumvent concerns that private plans are cherry picking healthy enrollees from public plans by performing our analyses at the state-level. Furthermore, we test for and fail to find changes in enrollee characteristics that are concurrent with privatization.

### C. *Alternative Empirical Strategies*

This section presents two additional alternative empirical strategies. First, one concern with our staggered difference-in-differences framework is that post-privatization changes in relative outcomes could be driven by changes in outcomes in control states. We evaluate this possibility by asking how likely it would be to observe breaks from trend in the control states. We create a placebo privatization date for each of the 16 control states by sampling with replacement from the weighted empirical distribution of privatization dates in the treatment states. We then estimate a regression that includes state fixed effects, and separate lags and leads for actual and placebo privatization. We repeat this exercise 1,000 times. Figure 10 plots the point estimates and 95 percent confidence interval of the effect of lags and leads of the actual privatization decision, the average of the 1,000 point estimates of the lags and leads of placebo treatment, and the fifth and ninety-fifth percentile of the 1,000 point estimates. The results of these analyses further illustrate that the estimated effects of privatization are driven by treatment states rather than by control states.[34]

Second, a related concern is that the effect sizes that we observe may be more likely than it would appear based on traditional statistical testing because of a small number of clusters.[35] Figure 11 presents a permutation test in which we randomly create 13 placebo privatization events in both treatment and control states. We then estimate the effect of placebo privatization on outcomes. We repeat this process 1,000 times. Figure 11 plots the distribution of the $\beta$ coefficients, compares the distribution to the central estimate, and lists how often the placebo treatments from the permutation tests result in an estimate more extreme than the one we actually observe. The distributions suggest that it is very unlikely that one would observe more extreme outcomes by chance.[36]

### D. *Medicare Falsification Exercise*

To show that changes in drug usage are not driven by contemporaneous state level factors, we present an additional placebo test using Medicare (Part D) data.[37]

[34] Online Appendix Figure A.21 presents results for additional dependent variables.

[35] Online Appendix Table A.31 illustrates that there are only small changes to statistical significance if we calculate $p$-values following the suggestions in Cameron, Gelbach, and Miller (2008).

[36] Online Appendix Figure A.22 presents results for additional dependent variables.

[37] This "placebo" group could be treated if privatization led to spillovers to other insurance segments, as identified by Lakdawalla and Yin (2015) for price negotiations in Part D.

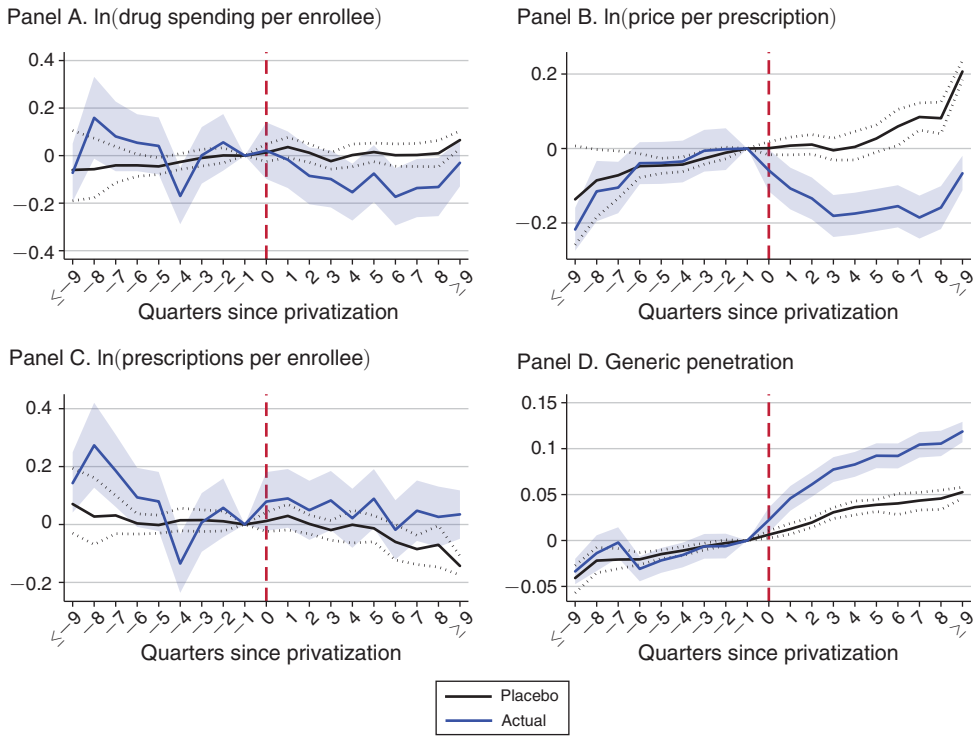


FIGURE 10. ACTUAL VERSUS PLACEBO PRIVATIZATION AND DRUG USAGE

*Notes:* Unreported controls include state fixed effects. Observations are weighted by state Medicaid drug spending in the second quarter of 2010. Point estimates for the effect of quarter pre-post privatization for actual privatization are presented as solid thin lines and 95 percent confidence intervals for those coefficients, based upon standard errors that are clustered by state, are presented with shading. Placebo effects are presented with a solid thicker line and the dash lines are the 95 percent confidence interval.

More specifically, we begin with a 10 percent sample of Medicare beneficiaries for 2010 through 2014. We restrict the sample to beneficiaries over 65 who never receive low-income subsidies. We recalculate each of our central measures for this sample.[38]

Figure 12 presents the relationship between lags and leads of the Medicaid privatization decision and Medicare Part D drug usage. Online Appendix Table A.30 presents pooled versions of this relationship. There is no evidence of any relationship between Medicaid privatization and *drug spending per enrollees*, days supply per enrollee (the analog of *prescriptions per enrollee*), point-of-sale price per days supply (the analog of *price per prescription*), *drug utilization per enrollee* or *generic efficiency* for Medicare Part D enrollees. The pooled estimate for *share high offset* is statistically significant, but the point estimate is small and results from a pre-trend. For *generic accessibility* (and therefore also *generic penetration*), there

[38] The measures that we construct for Medicare and Medicaid patients differ slightly because in the Medicaid data quantities are measured as prescriptions whereas in the Medicare data quantities are measured as days of drugs supplied.

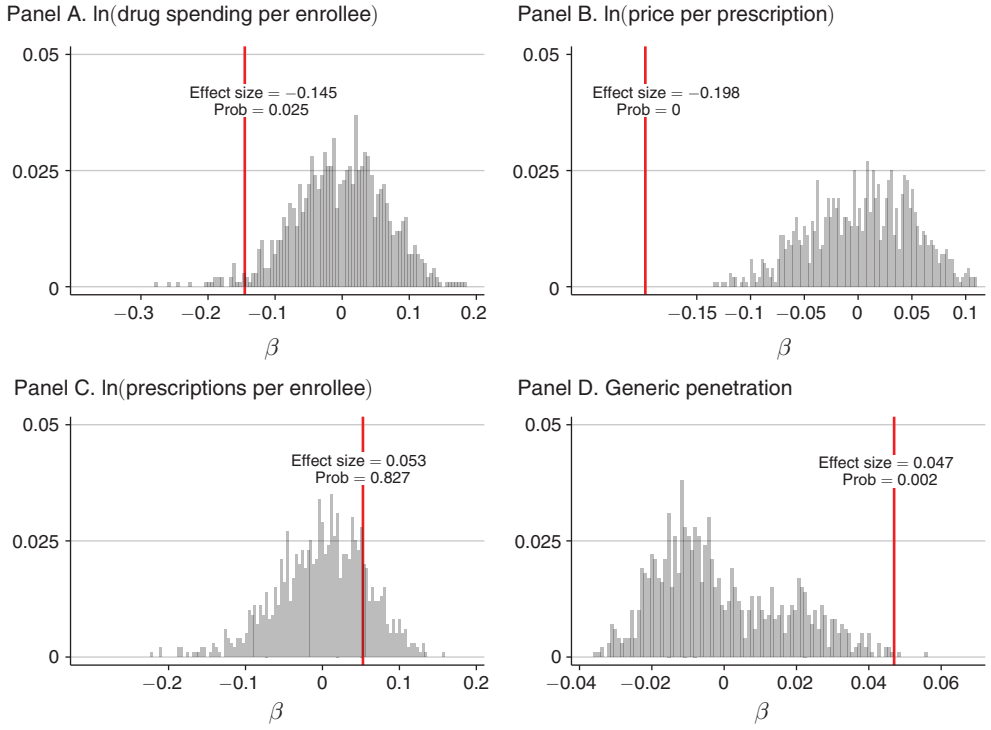


Figure 11. Distribution of Permutation Test Effect Sizes versus Actual Coefficient Estimates

*Notes:* We generated a new dataset under the null hypothesis that privatization had no effect. We reestimated the central regression for each variable and generate a new dataset in which we alter the data for the treatment states, so as to take out the estimated actual effect of privatization and proceed as described in the text.

is a noticeable break from trend at the time of privatization. The pooled estimate is statistically significant with a magnitude that is roughly one-fourth as large as that for the Medicaid drug sample. However, privatization's effect on *simulated generic accessibility* is nearly identical to its effect on *generic accessibility*, which means the increased use of generics in the Medicare Part D sample is driven exclusively by patent expiration.[39] Overall, the falsification exercise suggests that our results are not driven by concurrent state-level shocks.

## E. *Effect on Overall State Spending*

This section examines how privatization affects overall healthcare spending, and how states and MCOs split any savings. The overall effect of privatization will depend on the nature of procurement, the incentives the state creates for MCOs, and the level of competition in the insurance market.

[39] Figure 4 rules out this explanation for the Medicaid sample. The entry of generic Lipitor—used heavily in the Medicare population but not the Medicaid population of interest—explains a substantial share of increased generic accessibility among Medicare Part D patients.

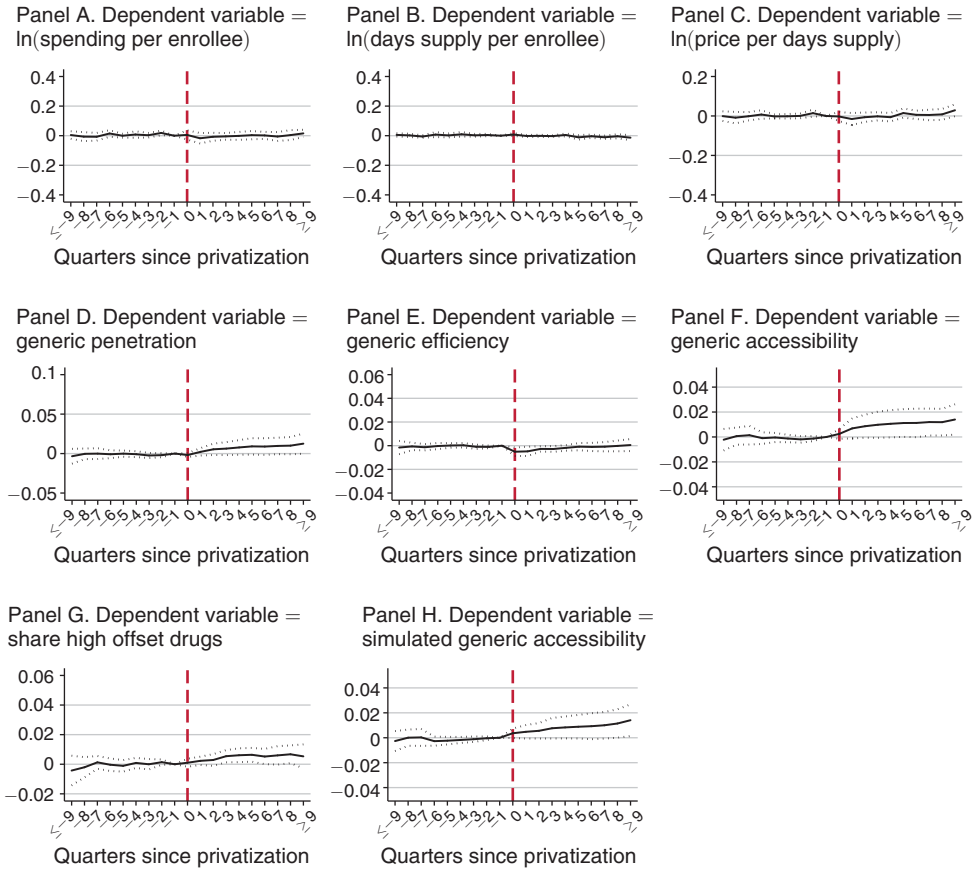


FIGURE 12. MEDICAID PRIVATIZATION AND MEDICARE DRUG USAGE (FALSIFICATION)

*Notes:* Unreported controls include state fixed effects, quarter fixed effects, and a post-Medicaid expansion indicator. Observations are weighted by state Medicaid drug spending in the second quarter of 2010. The solid line is point estimates for the effect of quarter pre-post privatization and the dotted lines are the 95 percent confidence interval for those coefficients, based upon standard errors that are clustered by state. Spending is computed for Medicare enrollees 65+ years old who never receive a low-income subsidy. Data covers the second quarter of 2010 through the fourth quarter of 2014.

We present multiple pieces of evidence that suggest how Medicaid drug privatization affects overall healthcare spending of Medicaid patients. First, an issue with interpreting our results is that they may not represent gains from privatization, but instead gains from having the same provider of medical and drug benefits. If this was the case, we would observe larger effects in the states that began with private medical benefits. We do not. A separate concern is that reductions in drug spending may lead to increases in medical spending. Three facts should alleviate this concern. First, in all of the states that privatize drug benefits, medical benefits are managed by the same MCO, so the MCO's incentives should be well-aligned. Second, privatization does not reduce the use of drugs with high medical offsets. Finally, Appendix E does not find any evidence that privatization affects emergency department use by Medicaid patients, although the estimates are imprecise.

Next, we present evidence that suggests that states capture at least part of the savings from lower drug spending. First, Appendix D examines the effect of privatization on total state spending on Medicaid, which depends on both drug spending and any spillover effects on other types of spending, as well as on how MCOs and states split any surplus that MCOs create. The estimates are noisy, but we are able to rule out large increases in spending. While we do not find evidence of changes in overall spending, the literature on offsets and recent work by Lavetti and Simon (2018) and Starc and Town (2015) highlight externalities to nondrug medical utilization. Next, Appendix F examines the effect of privatization on the difference between MCO premium revenues and claims costs. There is suggestive evidence (although not statistically significant at conventional levels) that privatization did increase MCO profits, but that this was driven by the expansion in demand (quantity/covered lives), rather than by higher margins. While we are unable to rule out the possibility that either the MCO or the state capture the full cost savings, we note that if MCOs were capturing the full surplus, then the difference between their premium revenues and costs would be higher.

Anecdotal evidence supports this interpretation. First, bidding for MCO contracts tends to be competitive: in Illinois, state officials only accepted six of nine bids.[40] When Iowa expanded its program from a single carrier to multiple carriers, it gave contracts to only four of ten bidders.[41] Second, states increasingly require plans to submit utilization information and incorporate that information into future reimbursements (Medicaid and CHIP Payment and Access Commission 2011); political scientists note that we should expect a ratchet effect compressing reimbursements each time the states rebid the contract (Thompson and Dilulio 1998).[42] Finally, minimum loss ratio regulations now require Medicaid MCOs to spend at least 85 percent of premium dollars on medical expenses, ensuring that the surplus is split between the parties.

## VII. Conclusion

State and federal officials have proposed a range of policies to combat rising drug prices. Many advocate for direct price negotiation between the Medicare program and drug manufacturers. This debate overlooks the fact that private MCOs administer a large percentage of both Medicare and Medicaid drug benefits; it is important to know whether outsourcing to MCOs has been successful. The policy change we study could save billions of dollars: privatizing 100 percent of the Medicaid drug benefits would reduce drug spending by 21.3 percent. In the 13 treated states, privatization reduced annual Medicaid drug spending by $1.2 billion.

Several prior studies compare the performance of publicly versus privately administered Medicaid benefits (Duggan 2004; Duggan and Hayford 2013; Aizer, Currie, and Moretti 2007; Currie and Fahr 2005; Vabson 2017). Our approach goes

[40] See Illinois Department of Healthcare and Family Services (2017). Nearly all states currently have multiple carriers.

[41] See Iowa Department of Human Services (2015).

[42] Our effects are consistent with Al-Ississ and Miller (2013), who find evidence of abnormal returns in the pharmaceutical sector after Scott Brown's election endangered passage of the ACA.

inside the "black box," identifies the sources of MCO savings, and considers possible harmful consequences. MCOs achieve one-third of savings by negotiating lower prices with pharmacies, and achieve two-thirds of savings by restricting formularies to steer demand to generics and drugs with high medical offsets. These results suggest that MCOs cut spending with a scalpel rather than a hatchet.

The pricing results are also consistent with those of Duggan and Hayford (2013). They find that privatization only reduces spending in states where administered prices are high. Pre-ACA, public insurers received large, mandated rebates unavailable to private plans. As a result, privatization was not attractive. By contrast, public insurers pay pharmacies higher prices than those that private plans negotiated. Once the ACA equalized rebates, privatization became attractive. Public insurers do not face competitive constraints on provider reimbursement rates. Therefore, the impact of privatization on prices depends on how both market and public reimbursement rates are set. Of course, the changes we study could have broader consequences for pharmacy market structure and pharmaceutical innovation (Blume-Kohout and Sood 2013). This is an important direction for future work.

In our setting, states calculate pharmacy reimbursements based on expected costs. This type of reimbursement is typical of public insurers. Furthermore, both Medicare and Medicaid cover a wide range of care, without regard to cost effectiveness. In these ways, our setting is a microcosm of publicly administered healthcare in the United States. Public coverage and reimbursement rules often avoid steering patients, even based on cost effectiveness.

Can public payers limit coverage and successfully incentivize valuable care? The federal government recently denied the Massachusetts Medicaid program's request to restrict the state-administered formulary and pharmacy network. However, Medicare is experimenting with value-based payments reforms. Public systems in other countries make coverage decisions based on cost effectiveness. Thus, the performance of publicly administered care in the United States will remain a fluid issue and empirical question.

## REFERENCES

**Aizer, Anna, Janet Currie, and Enrico Moretti.** 2007. "Does Managed Care Hurt Health? Evidence from Medicaid Mothers." *Review of Economics and Statistics* 89 (3): 385–99.

**Al-Ississ, Mohamad M., and Nolan H. Miller.** 2013. "What Does Health Reform Mean for the Health Care Industry? Evidence from the Massachusetts Special Senate Election." *American Economic Journal: Economic Policy* 5 (3): 1–29.

**Alpert, Abby, Mark Duggan, and Judith K. Hellerstein.** 2013. "Perverse Reverse Price Competition: Average Wholesale Prices and Medicaid Pharmaceutical Spending." *Journal of Public Economics* 108: 44–62.

**Anderson, Gerard F., Uwe E. Reinhardt, Peter S. Hussey, and Varduhi Petrosyan.** 2003. "It's the Prices, Stupid: Why the United States Is So Different from Other Countries." *Health Affairs* 22 (3): 89–105.

**Blewett, Lynn A., Julia A. Rivera Drew, Risa Griffin, Kari C.W. Williams, and Daniel Backman.** 2018. "IPUMS Health Surveys: Medical Expenditure Panel Survey, Version 1.0." Integrated Public Use Microdata Series. https://ipums.org/projects/ipums-health-surveys/d071.v1.0 (accessed January 15, 2019).

**Blume-Kohout, Margaret E., and Neeraj Sood.** 2013. "Market Size and Innovation: Effects of Medicare Part D on Pharmaceutical Research and Development." *Journal of Public Economics* 97: 327–36.

**Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller.** 2008. "Bootstrap-Based Improvements for Inference with Clustered Errors." *Review of Economics and Statistics* 90 (3): 414–27.

**Cardona, Claire.** 2011. "Lower Medicaid Dispensing Fees May Pressure Pharmacies." *New York Times*, October 14. https://www.nytimes.com/2011/10/14/us/lower-medicaid-dispensing-fees-may-pressure-pharmacies.html.

**Carey, Colleen.** 2017. "Technological Change and Risk Adjustment: Benefit Design Incentives in Medicare Part D." *American Economic Journal: Economic Policy* 9 (1): 38–73.

**Centers for Medicare and Medicaid Services.** 2002–2015. "Expenditure Reports from MBES/CBES." https://www.medicaid.gov/medicaid/financial-management/state-expenditure-reporting-medicaid-chip/expenditure-reports-mbescbes/index.html (accessed April 2017).

**Centers for Medicare and Medicaid Services.** 2010–2014. "Part D Claims Data." https://www.cms.gov/Medicare/Prescription-Drug-Coverage/PrescriptionDrugCovGenIn/PartDData.

**Centers for Medicare and Medicaid Services.** 2010–2017a. "Enrollment Report." https://www.medicaid.gov/medicaid/managed-care/enrollment-report/index.html (accessed February 2019).

**Centers for Medicare and Medicaid Services.** 2010–2017b. "State Drug Utilization Data." https://www.medicaid.gov/medicaid/prescription-drugs/state-drug-utilization-data/index.html (accessed February 2017).

**Centers for Medicare and Medicaid Services.** 2013–2018. "Drug Utilization Review Annual Report." https://www.medicaid.gov/medicaid/prescription-drugs/drug-utilization-review/annual-reports/index.html (accessed May 2017).

**Chandra, Amitabh, Jonathan Gruber, and Robin McKnight.** 2010. "Patient Cost-Sharing and Hospitalization Offsets in the Elderly." *American Economic Review* 100 (1): 193–213.

**Cooper, Zack, Stuart V. Craig, Martin Gaynor, and John Van Reenen.** 2019. "The Price Ain't Right? Hospital Prices and Health Spending on the Privately Insured." *Quarterly Journal of Economics* 134 (1): 51–107.

**Currie, Janet, and John Fahr.** 2005. "Medicaid Managed Care: Effects on Children's Medicaid Coverage and Utilization." *Journal of Public Economics* 89 (1): 85–108.

**Curto, Vilsa, Liran Einav, Amy Finkelstein, Jonathan Levin, and Jay Bhattacharya.** 2019. "Health Care Spending and Utilization in Public and Private Medicare." *American Economic Journal: Applied Economics* 11 (2): 302–32.

**Cutler, David M., Mark McClellan, and Joseph P. Newhouse.** 2000. "How Does Managed Care Do It?" *RAND Journal of Economics* 31 (3): 526–48.

**Dranove, David, Christopher Ody, and Amanda Starc.** 2021. "Replication data for: A Dose of Managed Care: Controlling Drug Spending in Medicaid." American Economic Association [publisher], Inter-university Consortium for Political and Social Research [distributor]. https://doi.org/10.3886/E115927V1.

**Duggan, Mark.** 2004. "Does Contracting Out Increase the Efficiency of Government Programs? Evidence from Medicaid HMOs." *Journal of Public Economics* 88 (12): 2549–72.

**Duggan, Mark, and Tamara Hayford.** 2013. "Has the Shift to Managed Care Reduced Medicaid Expenditures? Evidence from State and Local-Level Mandates." *Journal of Policy Analysis and Management* 32 (3): 505–35.

**Duggan, Mark, and Fiona Scott Morton.** 2010. "The Effect of Medicare Part D on Pharmaceutical Prices and Utilization." *American Economic Review* 100 (1): 590–607.

**Duggan, Mark, Amanda Starc, and Boris Vabson.** 2016. "Who Benefits When the Government Pays More? Pass-through in the Medicare Advantage Program." *Journal of Public Economics* 141: 50–67.

**Einav, Liran, Amy Finkelstein, and Maria Polyakova.** 2018. "Private Provision of Social Insurance: Drug-Specific Price Elasticities and Cost Sharing in Medicare Part D." *American Economic Journal: Economic Policy* 10 (3): 122–53.

**Finkelstein, Amy, Matthew Gentzkow, and Heidi Williams.** 2016. "Sources of Geographic Variation in Health Care: Evidence from Patient Migration." *Quarterly Journal of Economics* 131 (4): 1681–1726.

**Fisher, Elliott S., David E. Wennberg, Thrse A. Stukel, Daniel J. Gottlieb, F.L. Lucas, and Étoile L. Pinder.** 2003. "The Implications of Regional Variations in Medicare Spending. Part 1: The Content, Quality, and Accessibility of Care." *Annals of Internal Medicine* 138 (4): 273–87.

**Geruso, Michael, Timothy Layton, and Daniel Prinz.** 2019. "Screening in Contract Design: Evidence from the ACA Health Insurance Exchanges." *American Economic Journal: Economic Policy* 11 (2): 64–107.

**Goodman-Bacon, Andrew.** 2018. "Difference-in-Differences with Variation in Treatment Timing." NBER Working Paper 25018.

**Healthcare Cost and Utilization Project (H-CUP).** 2006–2015. "HCUP Fast Stats—State Trends in Emergency Department Visits by Payer." H-CUP. https://www.hcup-us.ahrq.gov/faststats/StatePayerEDServlet?state1=AZ (accessed May 2017).

**Ho, Kate, and Robin S. Lee.** 2017. "Insurer Competition in Health Care Markets." *Econometrica* 85 (2): 379–417.

**Illinois Department of Healthcare and Family Services.** 2017. "State of Illinois Medicaid Managed Care Organization Request for Proposals." https://www.illinois.gov/hfs/info/MedicaidManagedCareRFP/Documents/2018-24-001_Illinois_Medicaid_MCO_RFP.pdf.

**Iowa Department of Human Services.** 2015. "MED-16-009 High Quality Health Care Initiative— Bid Proposals." https://dhs.iowa.gov/MED-16-009_Bidders-Library/Bid_Proposals.

**Kaiser Commission on Medicaid and the Uninsured.** 2012. *Medicaid Managed Care: Key Data, Trends, and Issues*. Washington, DC: Henry J. Kaiser Family Foundation.

**Kuziemko, Ilyana, Katherine Meckel, and Maya Rossin-Slater.** 2018. "Does Managed Care Widen Infant Health Disparities? Evidence from Texas Medicaid." *American Economic Journal: Economic Policy* 10 (3): 255–83.

**Lakdawalla, Darius, and Wesley Yin.** 2015. "Insurers' Negotiating Leverage and the External Effects of Medicare Part D." *Review of Economics and Statistics* 97 (2): 314–31.

**Lavetti, Kurt, and Kosali Simon.** 2018. "Strategic Formulary Design in Medicare Part D Plans." *American Economic Journal: Economic Policy* 10 (3): 154–92.

**Levinson, Daniel R.** 2015. *Medicaid Rebates for Brand-Name Drugs Exceeded Part D Rebates by a Substantial Margin*. Washington, DC: Department of Health and Human Services Office of Inspector General.

**Medicaid and CHIP Payment and Access Commission (MACPAC).** 2011. *Report to Congress: The Evolution of Managed Care in Medicaid*. Washington, DC: MACPAC.

**Medicaid and CHIP Payment and Access Commission (MACPAC).** 2018. *Medicaid Payment for Outpatient Prescription Drugs*. Washington, DC: MACPAC.

**National Association of Insurance Commissioners.** 2007–2014. "Insurance Data Products." (accessed November 27, 2015)

**Ross, Jim.** 2017. "Small Pharmacies Fight to Stay in Medicaid." *Ocala StarBanner*, April 2. https://www.ocala.com/news/20170402/small-pharmacies-fight-to-stay-in-medicaid.

**Ruggles, Steven, Sarah Flood, Ronald Goeken, Josiah Grover, Erin Meyer, Jose Pacas, and Matthew Sobek.** 2018. "IPUMS USA: Version 8.0." Integrated Public Use Microdata Series, Minneapolos, MN. https://usa.ipums.org/usa/cite.shtml (accessed January 15, 2019).

**Starc, Amanda, and Ashley Swanson.** 2018. "Preferred Pharmacy Networks and Drug Costs." NBER Working Paper 24862.

**Starc, Amanda, and Robert J. Town.** 2015. "Externalities and Benefit Design in Health Insurance." NBER Working Paper 21783.

**Thompson, Frank J., and John J. Dilulio.** 1998. *Medicaid and Devolution: A View from the States*. Washington, DC: Brookings Institution Press.

**Town, Robert, and Su Liu.** 2003. "The Welfare Impact of Medicare HMOs." *RAND Journal of Economics* 34 (4): 719–36.

**US Food and Drug Administration.** 2000–2016a. "Drugs@FDA: FDA-Approved Drugs." US Food and Drug Administration. https://www.accessdata.fda.gov/scripts/cder/daf/ (accessed February 2017).

**US Food and Drug Administration.** 2000–2016b. "National Drug Code Directory." US Food and Drug Administration. https://www.fda.gov/drugs/drug-approvals-and-databases/national-drug-code-directory (accessed February 2017).

**Vabson, Boris V.** 2017. "Efficiency Gains under Incomplete Contracting: Evidence from Medicaid." Stanford Institute for Economic Policy Research Job Market Paper.

Copyright of American Economic Journal: Applied Economics is the property of American Economic Association and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.